

BRAC University | June 2026

DEEP LEARNING FRAMEWORKS FOR SUSPICIOUS ACTIVITY DETECTION IN VIDEO SURVEILLANCE SYSTEMS



Research Overview

Deep Learning Frameworks for Suspicious Activity Detection in Video Surveillance Systems

Focus

- Video Anomaly Detection
- Intelligent Surveillance
- Public Safety Monitoring
- Automated Incident Response

Objective

- Detect suspicious activities automatically from surveillance videos

Research Motivation

Why This Research?

Limitation of Existing Studies

- Models evaluated on isolated datasets
- Difficult to compare architectures fairly
- Limited assessment of real-world generalization

Our Solution

- Unified and comprehensive evaluation framework
- Multi-dataset benchmark for comprehensive comparison

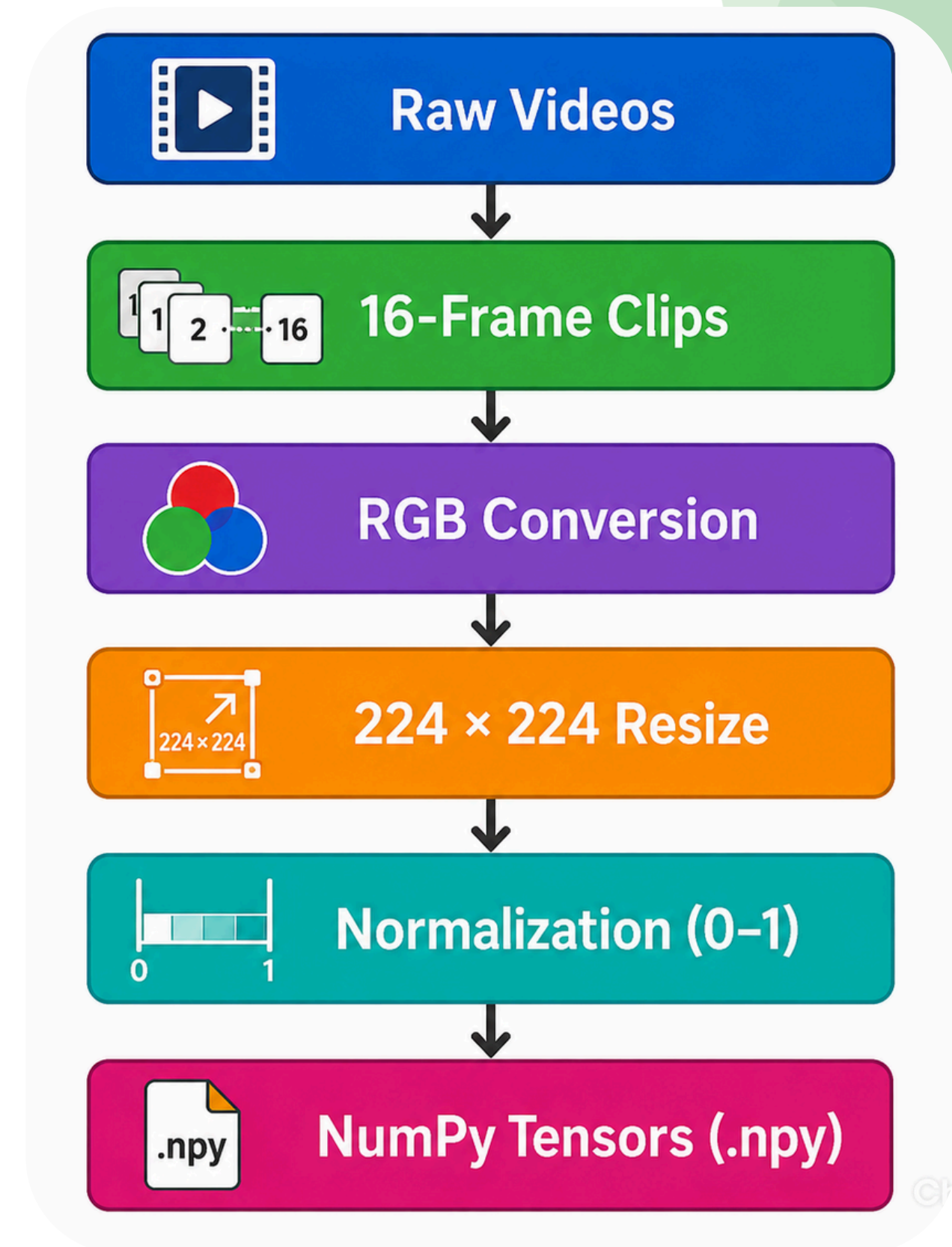
Dataset	Total Videos	Anomaly Videos	Normal Videos	Primary Scenario
UCF-Crime	1,900	950	950	Real-world surveillance crimes
XD-Violence	4,754	2,405	2,349	Multi-scene violence detection
RWF-2000	2,000	1,000	1,000	Real-world fight surveillance
UBI-Fights	1,000	216	784	Outdoor fight scenarios
SCVD	300	150	150	Campus violence detection
VAD_Binary (Total)	9,954	4,721	5,233	Merged benchmark

VAD_Binary Dataset Construction

- Unified multiple benchmark datasets
- Binary classification:
 - Class 0: Normal
 - Class 1: Anomaly
- Encourages learning of generalized abnormal behaviors

Key Benefits

- Standardized input format
- Reduced computational overhead
- Consistent benchmarking environment



Research Framework

Dataset Partitioning

- Training: 70%
- Validation: 15%
- Testing: 15%

Stratified Sampling

- Fixed random seed: 42
- Preserves anomaly-to-normal ratio across all subsets

Class Imbalance Mitigation

- Real-world surveillance is heavily normal-dominated
- Computed class weights from training labels
- Penalized majority-class bias during learning

Training Data Format

- Tensor Shape: $16 \times 224 \times 224 \times 3$
- Batch Size: 4

Data Augmentation Strategies

Spatial Augmentation

- Horizontal Flip
- Brightness Adjustment
- Contrast Adjustment
- Applied consistently across all 16 frames

Video-Aware Mixing Techniques

MixUp

- Blend two video clips and labels

CutMix

- Insert spatial patch across all frames
- Adjust label using mixed-region ratio

Four Augmentation Settings

Strategy	Purpose
No Augmentation	Baseline performance
Spatial Augmentation	Improve visual robustness
MixUp	Smoother decision boundaries
CutMix	Better feature localization

Why?

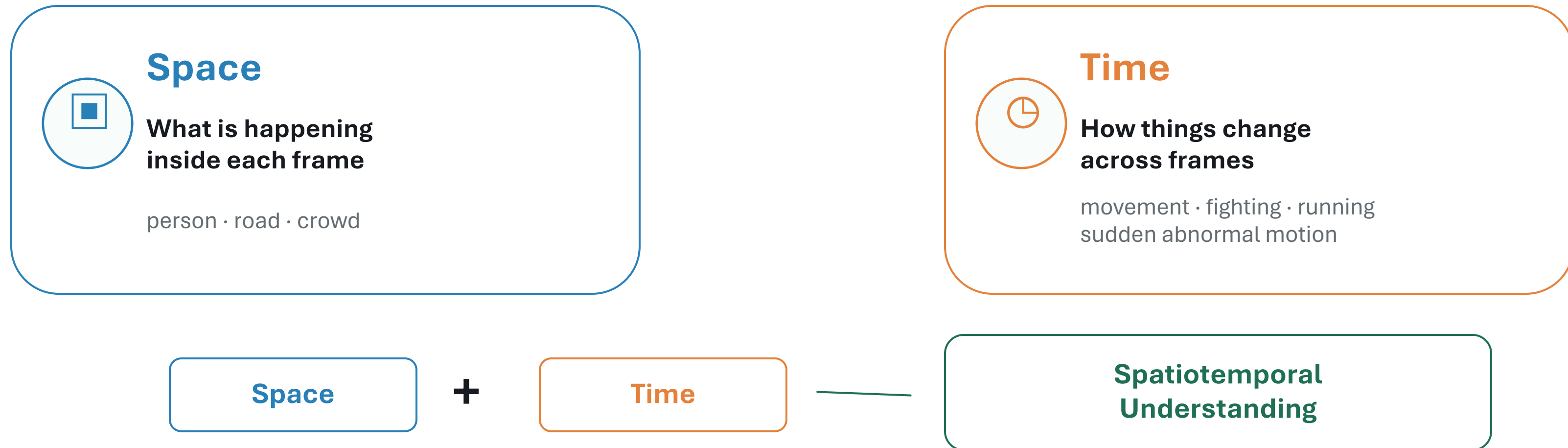
- To preserve the physical realism and lighting of the scene
- To be much more reliable on edge cases
- To preserve the temporal context

Efficient Training

- Augmentations applied on-the-fly
- Custom data generator
- No additional storage overhead

Nano Family Models

Video anomaly detection must understand both space and time.

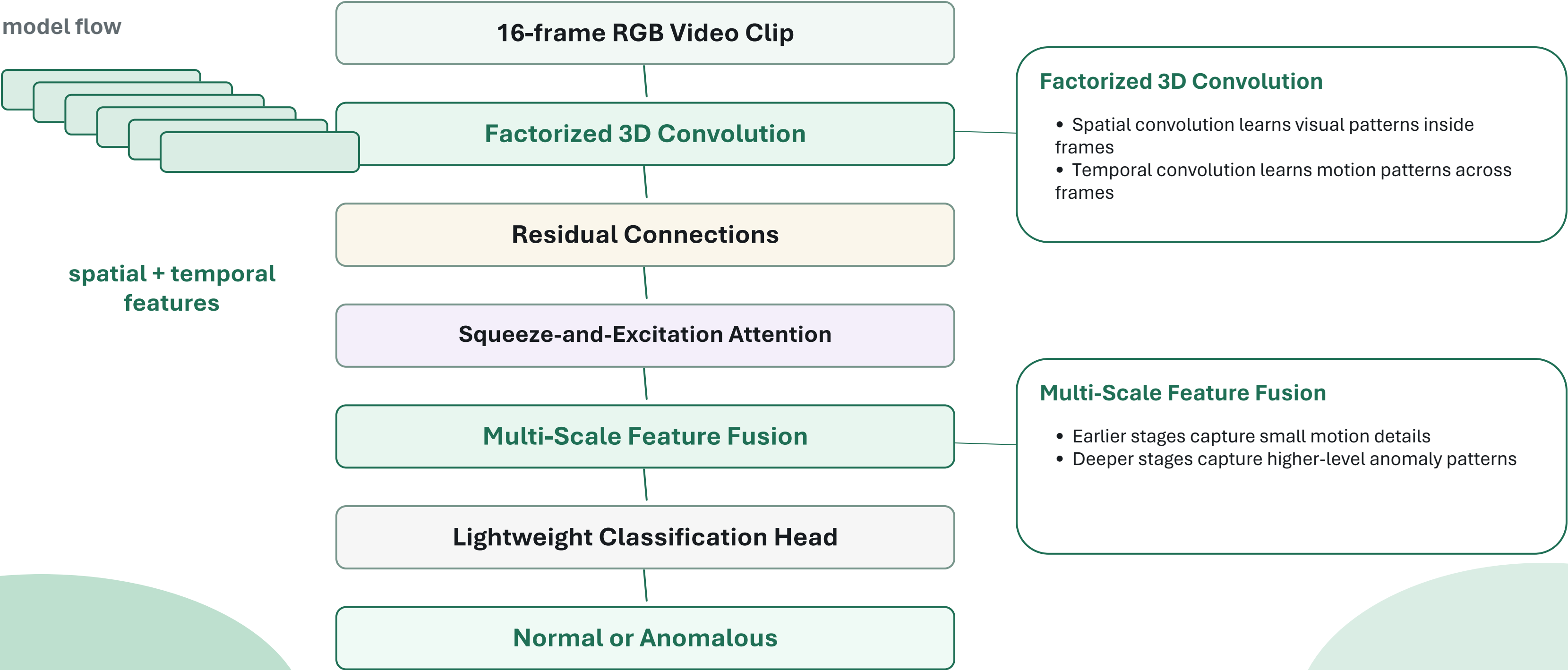


Traditional 3D CNNs can capture spatiotemporal information, but many are computationally heavy.

That is why we proposed the Nano3D family.

Nano3D: Balanced Compact Model

Compact multi-scale spatiotemporal model for binary anomaly detection



Nano3D Family Variants

Nano3D



Balanced version

- ✓ Compact multi-scale spatiotemporal model
- ✓ Best practical balance

Nano3D Edge



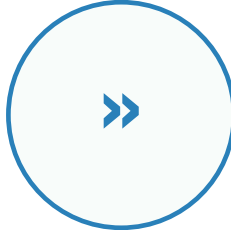
Ultra-lightweight version

16 thousand parameters

0.06 MB model size

- ✓ Designed for edge devices
- ✓ Useful when memory and computation are very limited

Nano3D Turbo



Speed-oriented version

- ✓ Designed for faster video processing
- ✓ Grouped 3D convolutions
- ✓ Aggressive downsampling
- ✓ Focused on speed and parallel processing

Nano3D = balanced

Nano3D Edge = ultra-lightweight

Nano3D Turbo = speed-oriented

Why Nano3D is Practical

A surveillance model should not only be accurate, but also fast and lightweight enough for practical deployment.

Heavy 3D CNNs

C3D / I3D / ResNet3D

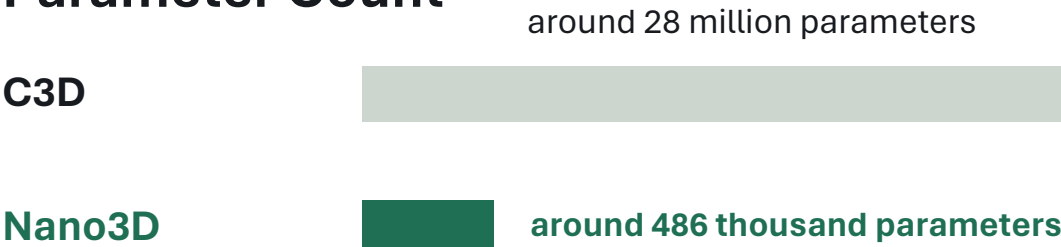
- Large 3D convolutional structures
- Higher parameter count
- Larger model size
- Higher inference cost

VS

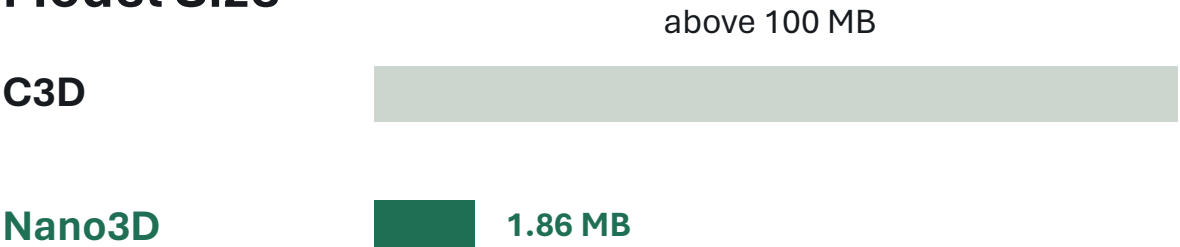
Nano3D

- Around 486 thousand parameters
- 1.86 MB model size
- Compact architecture
- Better practical trade-off

Parameter Count



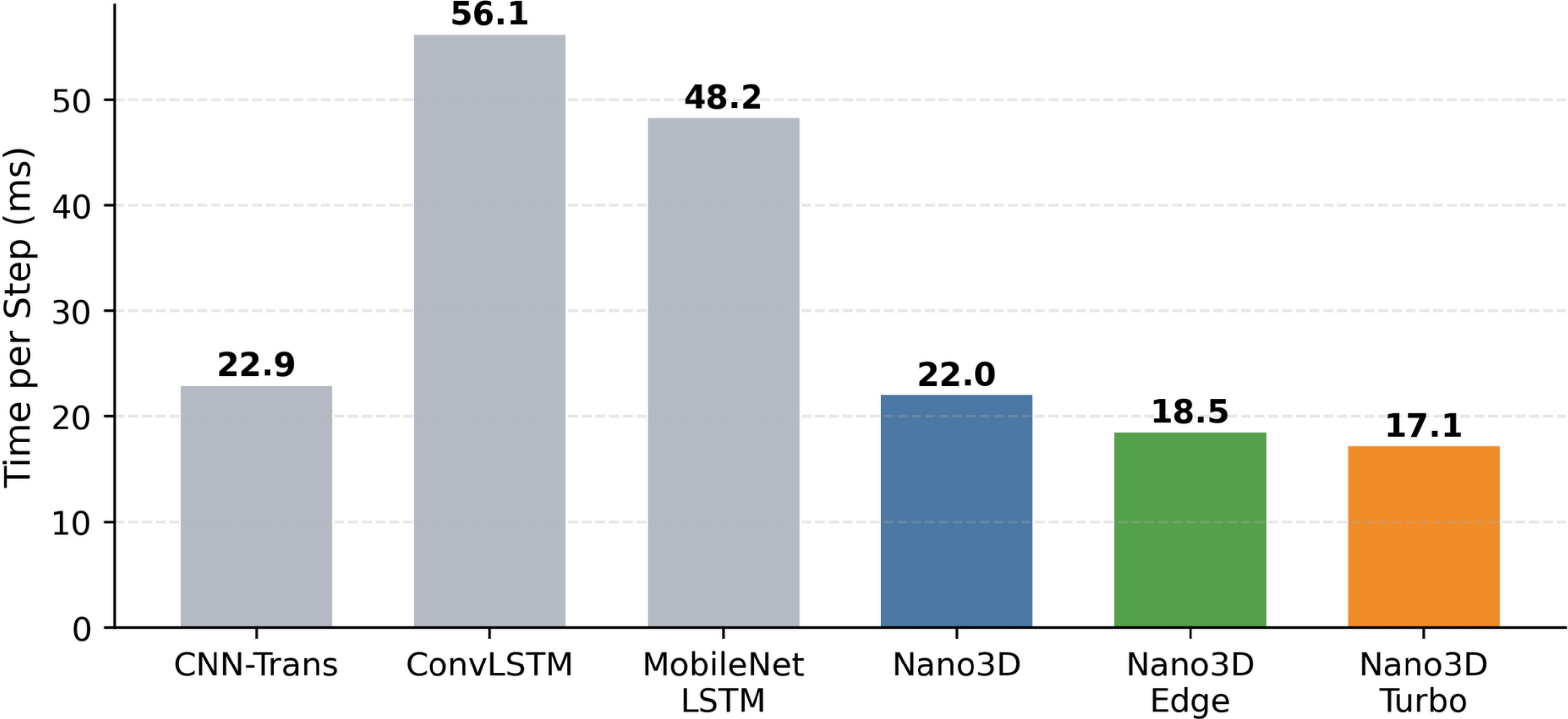
Model Size



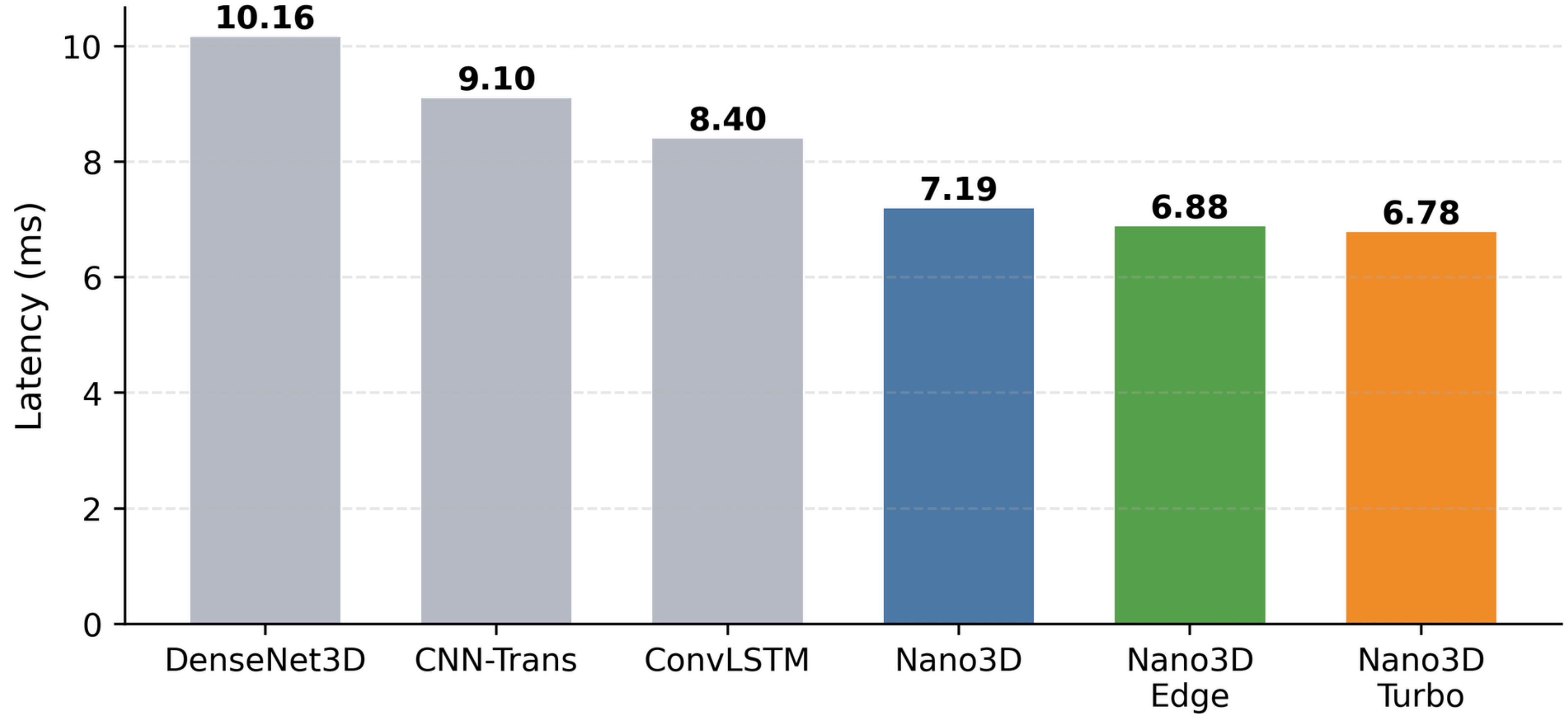
A carefully designed compact architecture can provide a better practical trade-off between accuracy, efficiency, and deployability.

Model	Total Params	Trainable Params	Non-trainable Params	Model Size (FP32)
ResNet3D	33.22 M	33.21 M	9.6 K	126.72 MB
I3D	4.51 M	4.50 M	6.5 K	17.19 MB
DenseNet3D	1.38 M	1.36 M	16.2 K	5.26 MB
ConvNeXt3D	5.87 M	5.87 M	0	22.39 MB
ResNeXt3D	1.42 M	1.41 M	9.7 K	5.43 MB
EffNet3D	986 K	981 K	5.0 K	3.76 MB
ViViT	540 K	540 K	0	2.06 MB
SlowFast	138 K	135 K	2.9 K	0.53 MB
R(2+1)D	15.39 M	15.37 M	17.4 K	58.69 MB
X3D-M	1.75 M	1.74 M	11.7 K	6.67 MB
C3D	28.19 M	28.18 M	5.5 K	107.54 MB
CNN-Trans	981 K	980 K	1.2 K	3.74 MB
ConvLSTM	2.91 M	2.91 M	2.0 K	11.09 MB
MobileNet-LSTM	3.86 M	1.61 M	2.26 M	14.74 MB
ResNet50-Attn	40.91 M	17.32 M	23.59 M	156.04 MB
Dense-BiLSTM	9.73 M	3.76 M	5.96 M	37.10 MB
Nano3D	486 K	484 K	2.3 K	1.86 MB
Nano3D-Edge	16 K	15 K	1.0 K	0.06 MB
Nano3D-Turbo	197 K	194 K	3.0 K	0.75 MB

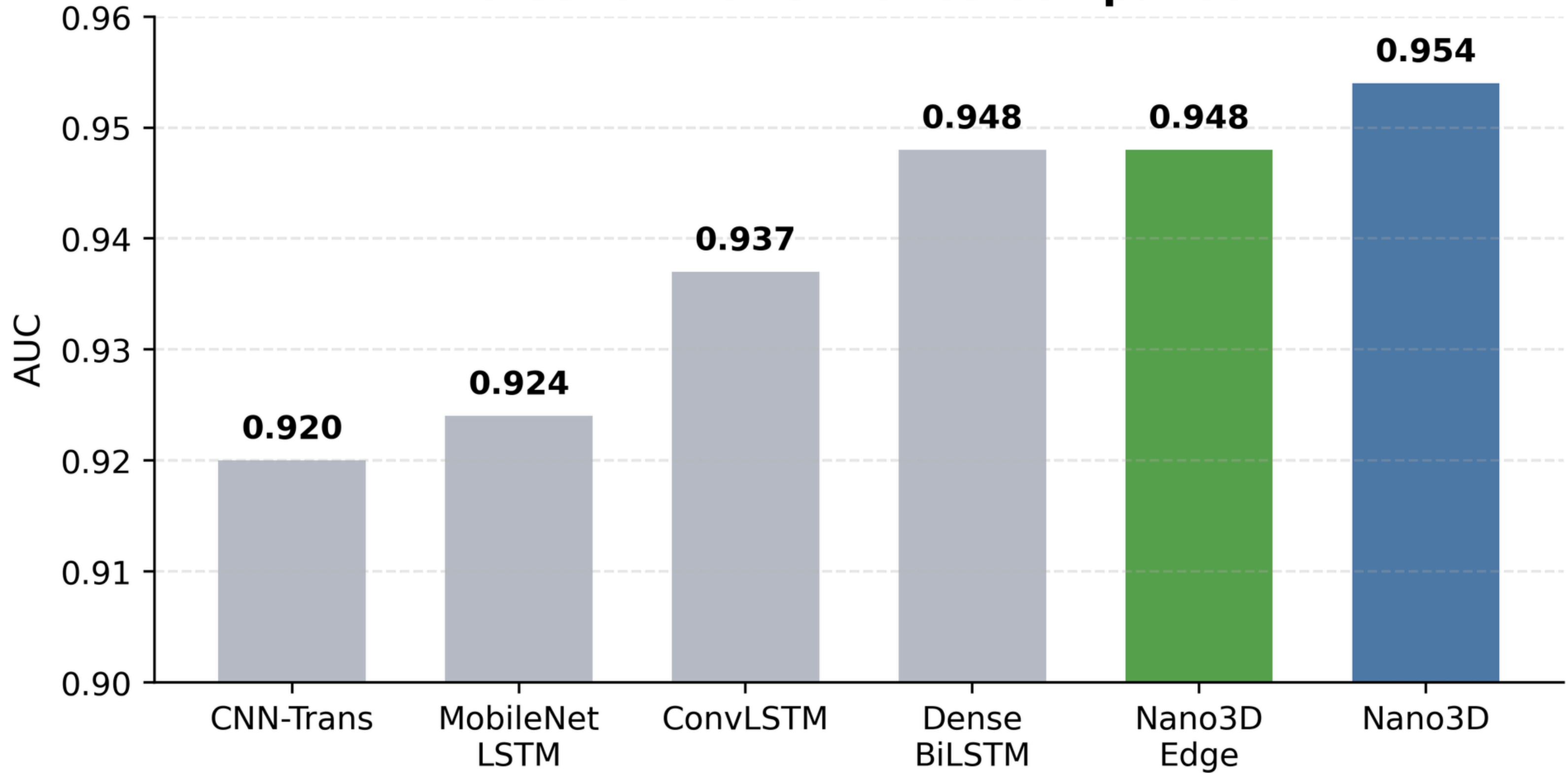
Training Efficiency Comparison (Lower is Better)



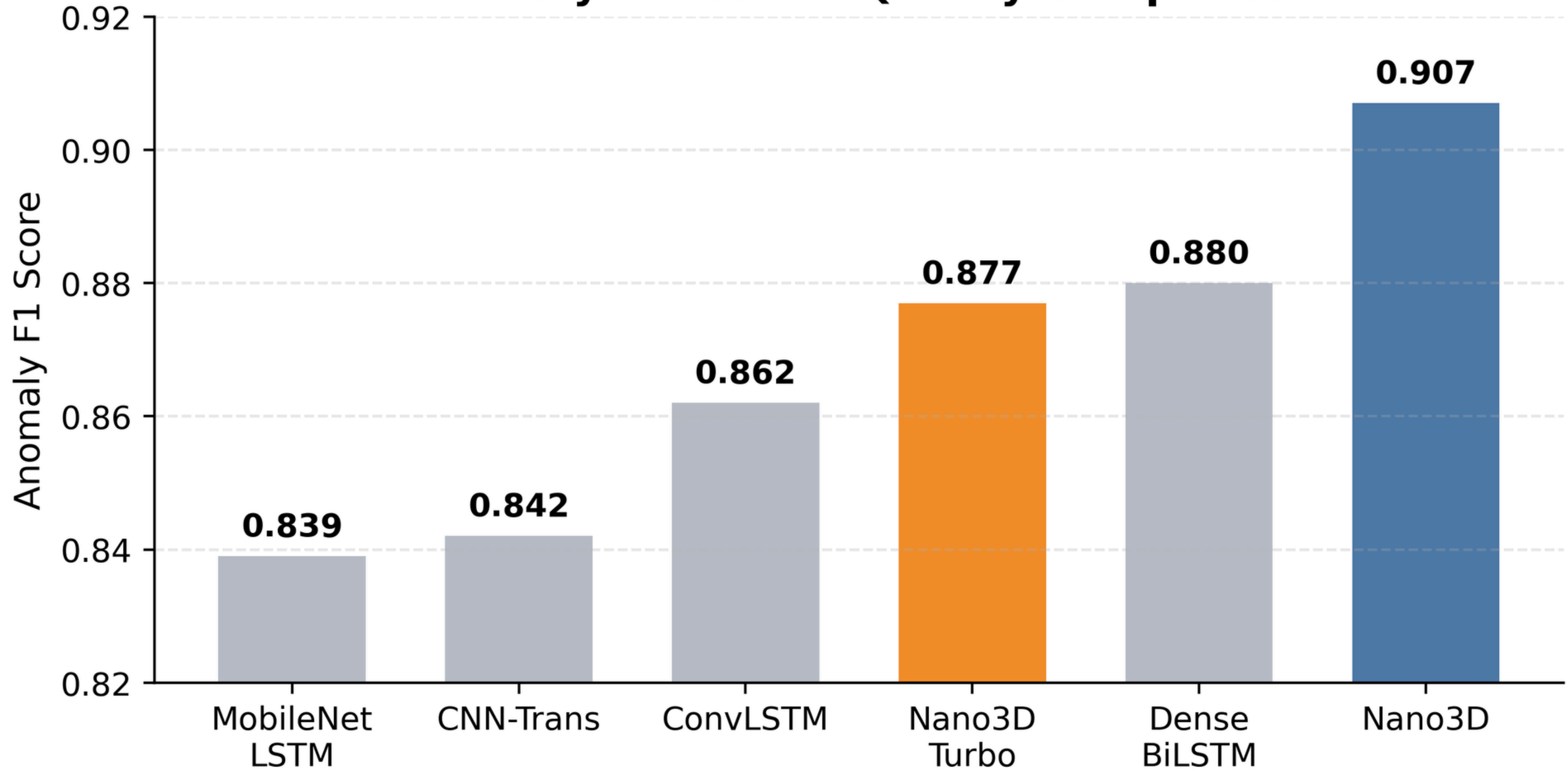
Inference Latency Comparison (Lower is Better)



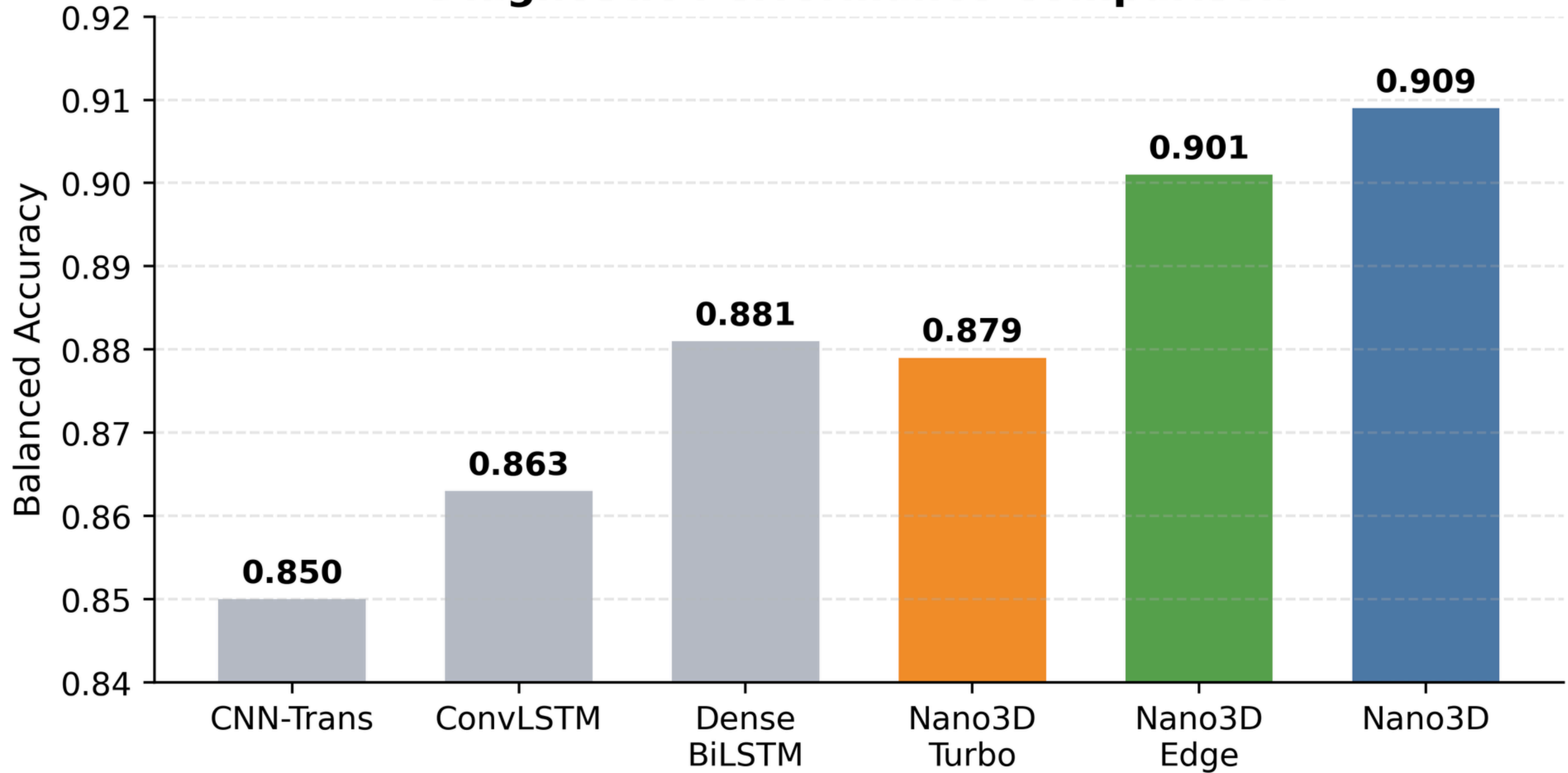
Detection Performance Comparison



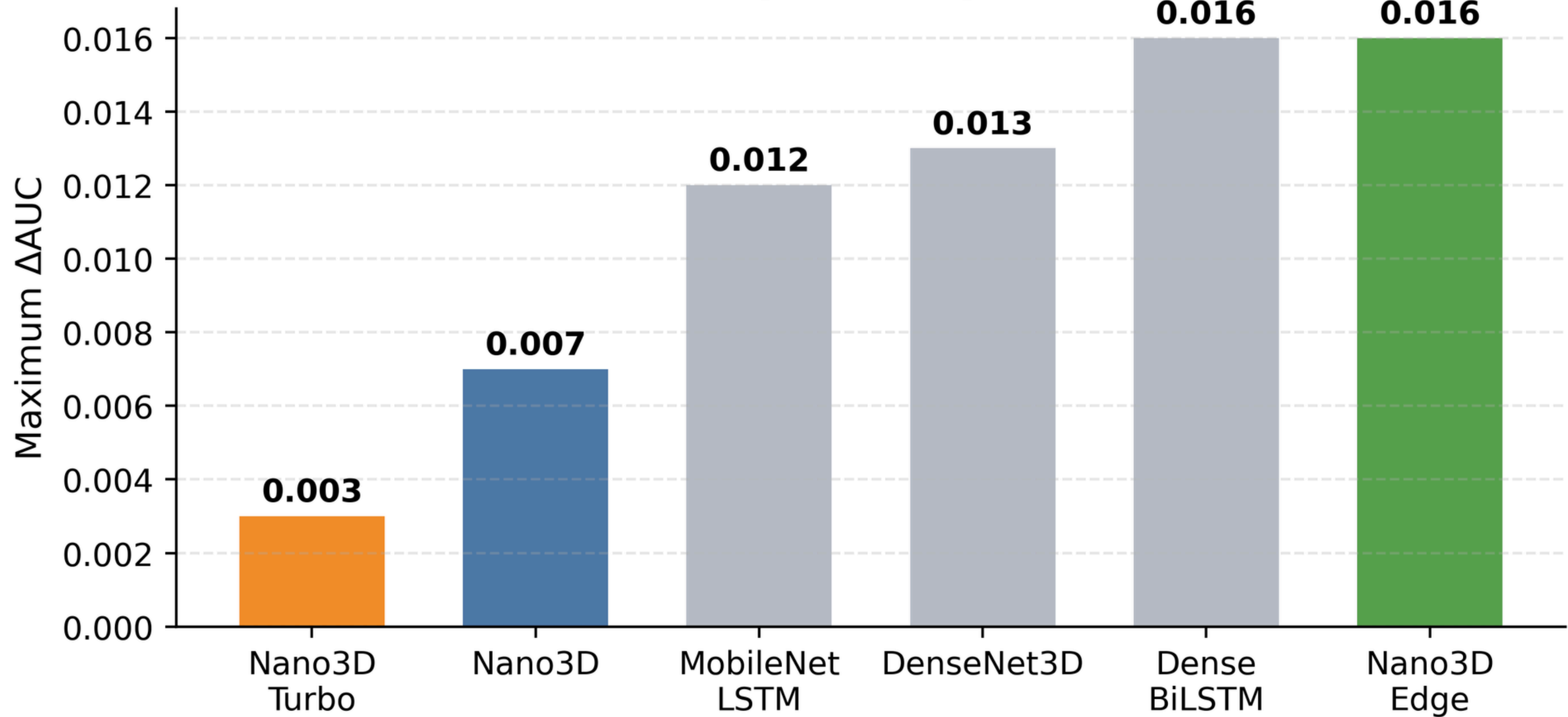
Anomaly Detection Quality Comparison



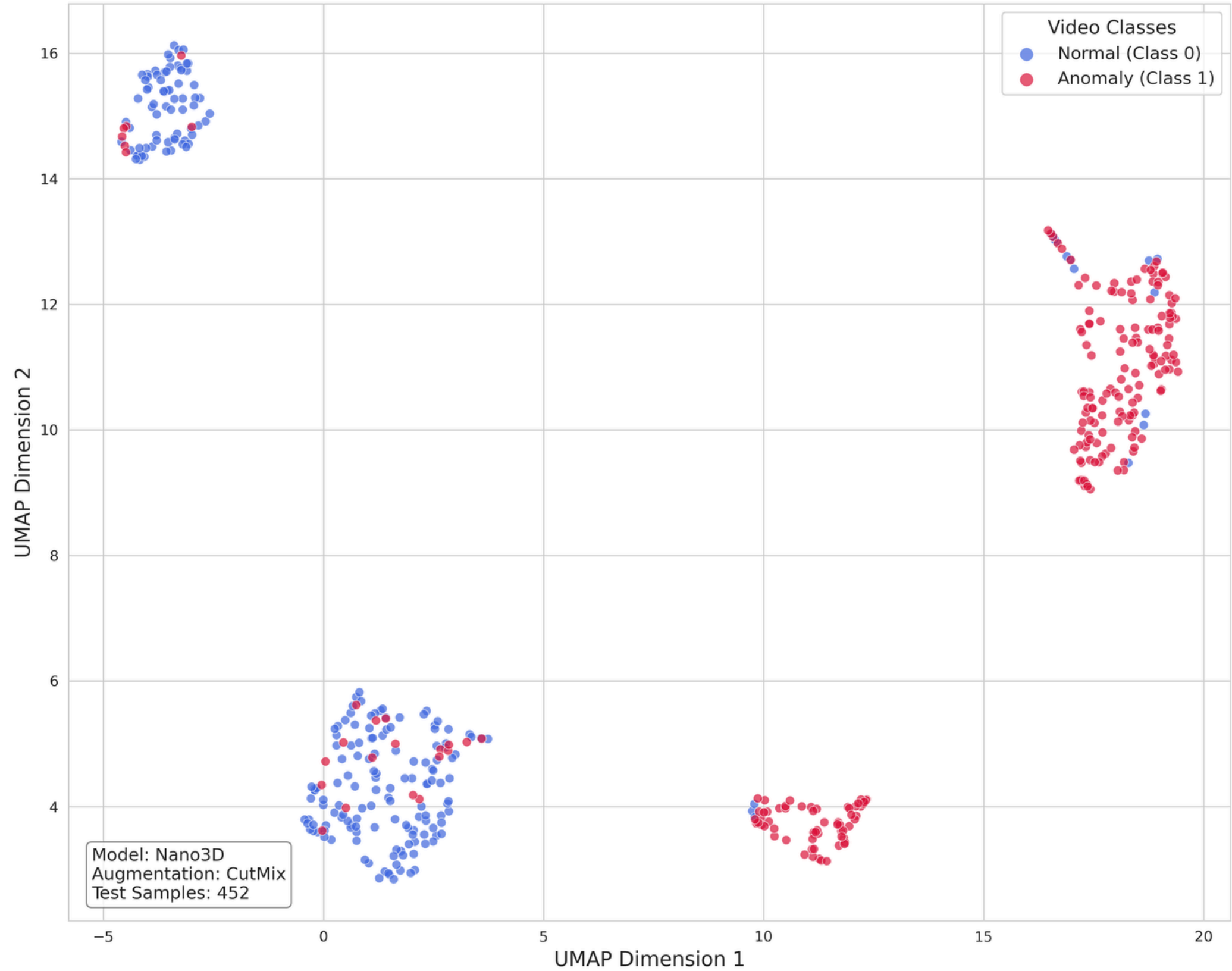
Diagnostic Performance Comparison



Model Robustness to Data Augmentation (Lower Dependency is Better)

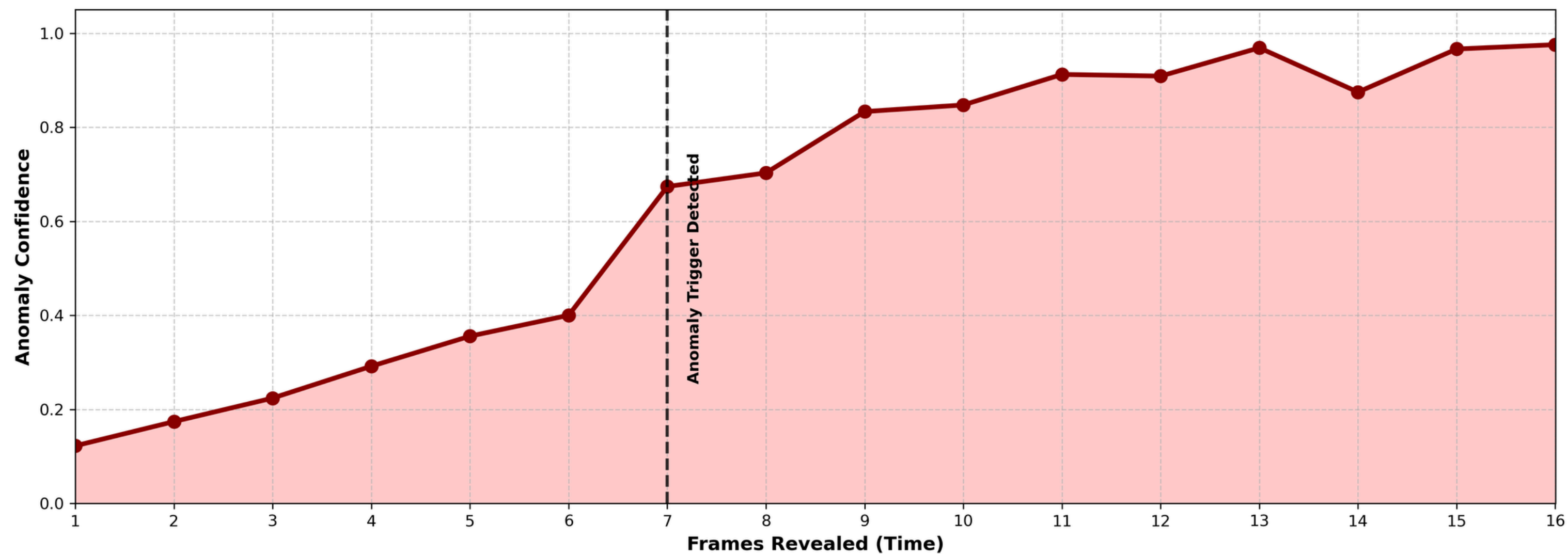


UMAP Latent Space Visualization of Nano3D Embeddings



Temporal Confidence Trajectory | Nano3D (CutMix)

Final Anomaly Score: 0.9758



Frame 1
Score: 0.12



Frame 6
Score: 0.40



Frame 11
Score: 0.91



Frame 16
Score: 0.98



Causal Perturbation | Nano3D (CutMix)

Score Drop: 0.9803 $\rightarrow$ 0.9711



3D Score-CAM | Nano3D (CutMix) | Anomaly Confidence: 0.9776

Frame 1



Score-CAM Overlay



Frame 6



Score-CAM Overlay



Frame 11



Score-CAM Overlay



Frame 16



Score-CAM Overlay



Spatiotemporal Grad-CAM | Nano3D (CutMix) | Anomaly Confidence: 0.9826

Frame 1



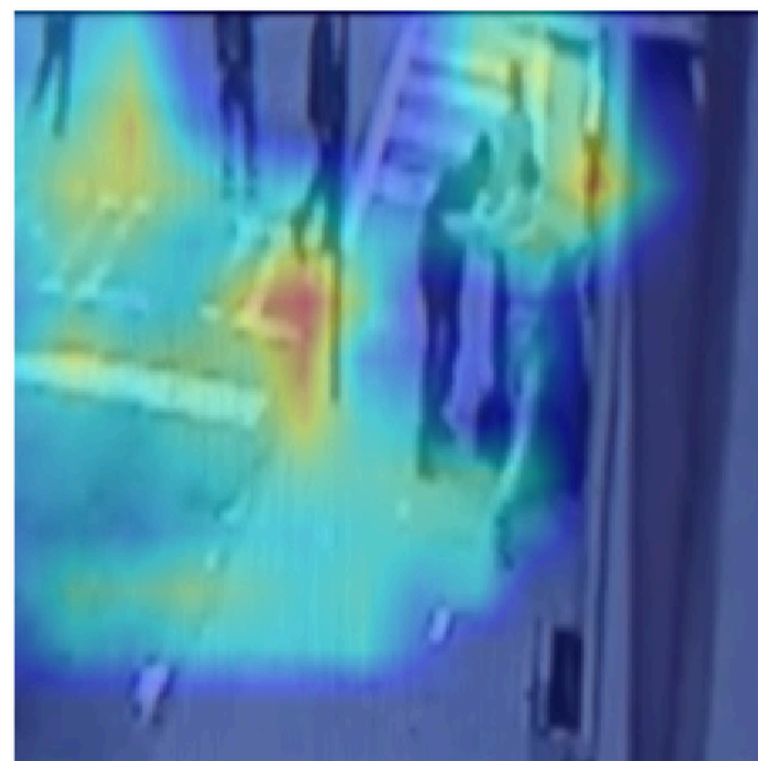
Grad-CAM Activation



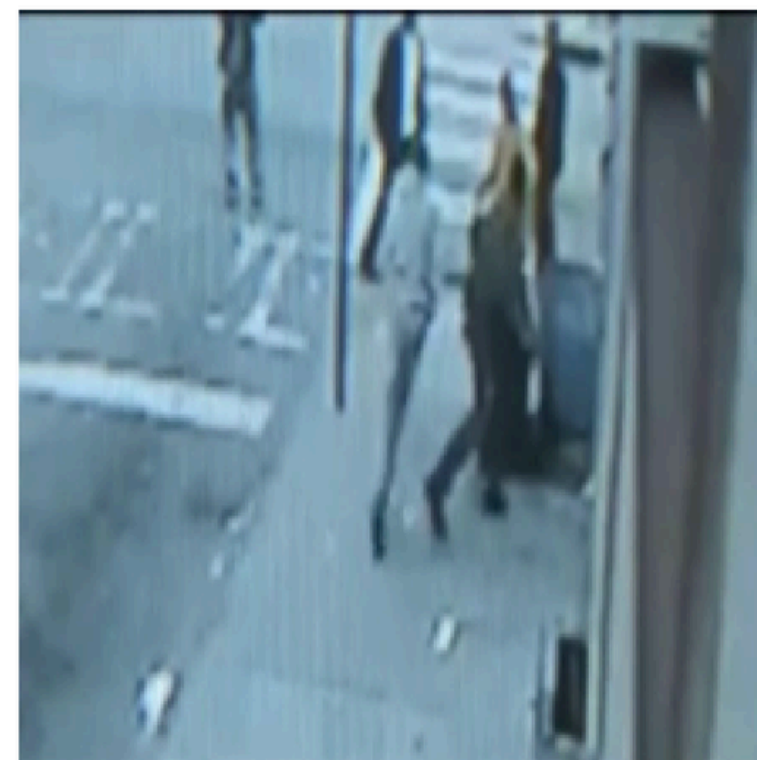
Frame 6



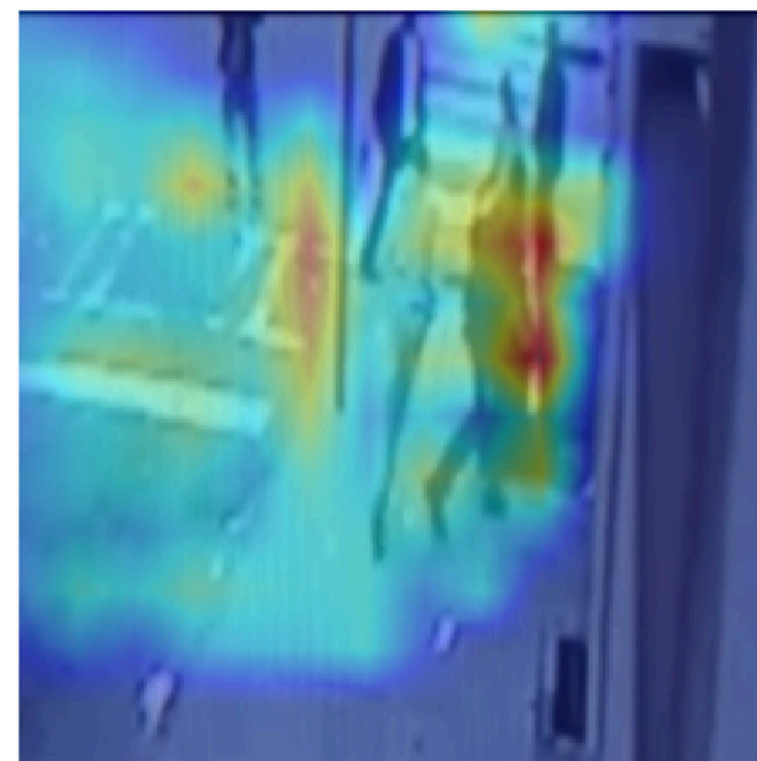
Grad-CAM Activation



Frame 11



Grad-CAM Activation



Frame 16



Grad-CAM Activation



The background of the slide features abstract, flowing, wavy shapes in various shades of green, ranging from light mint to a deeper forest green. These shapes are layered and overlap, creating a sense of movement and depth. The central area of the slide is a plain white background where the text is located.

Questions?